First assignment

Product Engineer
Priyanshu Anand

1. Candidate details

Field	Details
Name	Priyanshu Anand
Phone	+91 6205146659
Email	priyanshu82711@gmail.com
LinkedIn	https://www.linkedin.com/in/priyanswho/
GitHub	github.com/priyanshwho

2. Code change

Proposed commit message

```
Return -1 for non-integer inputs to multiplyByFive

Check is_int before multiplying. Preserve multiplication for
integer inputs and return -1 for other input types.
```

Concerns and alternative

The function now rejects numeric strings and floats that previously could be multiplied. That is a behavior change for existing callers. The meaning of `-1` must also be documented and checked by every caller.

I would first confirm which input types the function should accept. If only integers are valid, I would prefer an integer parameter and strict typing at the caller, or explicit validation that throws `InvalidArgumentException`. This makes misuse visible instead of returning an ordinary integer sentinel.

The guard does not prevent overflow. A large integer multiplied by five can become a float, conflicting with the integer return type. I would validate the supported range and test zero, negatives, numeric strings, floats, null, and integer limits.

3. Task approach

1. **Clarify the outcome.** Ask Sales who should subscribe, why they would want the newsletter, where traffic comes from, and how Sales will judge success. Define a completed signup before choosing a metric.
2. **Agree the minimum scope.** Confirm required fields, content, frequency, consent wording, confirmation behavior, brand assets, deadline, and the person who approves the page.
3. **Check the existing systems.** Identify the website, email platform or CRM, owner of the mailing list, and available integration. Confirm duplicate, unsubscribe, and data-deletion behavior.
4. **Write acceptance criteria.** A visitor can understand the offer, enter a valid email, consent, and get a truthful result. Include invalid input, duplicates, network errors, mobile layout, and keyboard access.
5. **Review a small prototype.** Share the content and signup flow with Sales. Resolve open questions and estimate delivery after the integration and approval needs are known.
6. **Build and verify.** Implement the page and server validation. Test the real integration in a sandbox if available. Check persistence, retries, accessibility, and approved analytics without recording email addresses in analytics.
7. **Release and hand over.** Get content approval, deploy through the existing release process, verify a signup, and monitor errors and the agreed conversion measure.

4. First communication to Sales

To: Sales team
Subject: Inputs needed for the newsletter signup page

Hi team,

I can start the newsletter signup page. To make the first version useful, could you help me confirm:

- Who is the audience, what will subscribers receive, and how often?
- Where will visitors find the page, and what result should we measure?
- Which fields do you need beyond email? Please share approved copy, brand assets, and an example if you have one.
- Which email platform or CRM should receive signups, and who can help with access and testing?
- Should signup require email confirmation? Who approves consent, privacy, and unsubscribe wording?
- What is the target launch date, and who gives final approval?

Could we have a 15-minute discussion tomorrow, or could you reply by tomorrow afternoon? Once we confirm the scope and integration, I will share a short implementation plan and delivery estimate. My proposed first version is an email field, explicit consent, clear validation, and a confirmation state.

Thanks,
Priyanshu Anand

5. UI and UX assessment

Reviewed shareatdoorstep.com/donate/ on 27 October 2026 in a desktop browser. Used the public Indiranagar, Bengaluru locality, selected Light Pickup, added one clothing item, and inspected scheduling. No donor identity, OTP, order, or payment was submitted.

#	Summary	Description	Type
1	Clear process outline	The five named steps show service selection, donation details, scheduling, verification, and placing an order. This helps donors anticipate the work.	Good
2	Location entry has alternatives	Manual search and Auto-Detect Location are both offered. The page explains that location is needed to check coverage.	Good
3	Useful service comparison	Light and heavy pickup show starting prices and capacity limits. A free self-drop option is also listed, giving donors a lower-cost alternative.	Good
4	Capacity feedback	Adding one clothing item changed the capacity display from 0% to 5% and the summary to one item. This makes packing constraints visible during selection.	Good
5	Cost appears after location	The initial location screen does not show a pickup price. Show an indicative fee range and mention self-drop before asking for a precise location.	Bad
6	Search depends on a placeholder	The address input is identified by “Building name, Block #, Area...” rather than a persistent field label. Add a visible label that remains after typing.	Bad
7	Support links need accessible names	The phone and WhatsApp links appear as icon glyphs in the accessibility tree. Give each a meaningful accessible name such as “Call support”.	Bad
8	Precise map confirmation	After selecting a locality, donors are asked to verify the pin before seeing services. This can improve routing, but a locality or postcode coverage check first would reduce initial effort.	Neutral

Suggested priority

First disclose the likely pickup cost and provide persistent labels. Then improve support-link names and let donors explore service availability before confirming an exact pickup point.

> I reviewed the flow up to scheduling. Scheduling displayed dated time slots, an optional pickup note, donor details, terms, and a tax-inclusive fee summary. Donor verification was not completed. No conclusions are made about checkout completion, delivery reliability, mobile behavior, or payment success.

6. Fundraiser listing schema

Reviewed milaap.org/crowdfunding/fundraisers and Refine Search on 27 October 2026. Cards show an image, title, percentage funded, amount raised, creator, and a fundraiser URL. Filters include cause categories, tax benefits, urgency, and donation matching.

> This is a proposed schema inferred from the public interface. The actual database structure is not public.

Table	Key fields	Purpose
users	id PK, display_name, created_at	Fundraiser creator.
fundraisers	id PK, creator_id FK, slug UNIQUE, title, story, goal_minor, currency, status, is_urgent, tax_benefit_eligible, matching_enabled, published_at	Campaign content and listing filters.
categories	id PK, slug UNIQUE, name	Medical, education, and other causes.
fundraiser_categories	fundraiser_id FK, category_id FK, composite PK	Many-to-many cause membership.
media	id PK, fundraiser_id FK, url, alt_text, position, is_cover	Campaign images and cover selection.
donations	id PK, fundraiser_id FK, donor_id nullable FK, amount_minor, refunded_minor, currency, status, payment_reference UNIQUE, created_at	Payment ledger and anonymous donations.

Relationships and correctness

A creator has many fundraisers. A fundraiser has many donations and media records, and may belong to several categories. Store money as integer minor units with an ISO currency code. Reject non-positive goals and donation amounts. Limit refunds to the original donation amount.

Calculate amount raised from captured donations minus refunds. Failed or pending payments do not count. Percentage funded is raised divided by goal, multiplied by 100. Permit overfunding in the stored value; the progress-bar width can be capped separately.

The browser displayed dollar amounts during this session. Currency display may differ from a campaign's settlement currency. Conversion needs an explicit exchange-rate source and timestamp. Never add mixed currencies directly.

Listing performance

Index published status and publication date with id as a tie-breaker. Use category join indexes and keyset pagination. A listing aggregate can cache raised totals if it is updated idempotently from verified payment events and reconciled with the ledger. Keep private donor information out of listing responses.